

## Cloud Computing Tentative Questions

### 1. What is Cloud Computing? What are the characteristics of Cloud Computing?

**Cloud Computing** is the delivery of computing services—including servers, storage, databases, networking, software, analytics, and intelligence—over the Internet ("the cloud"). It allows users to access computing resources on demand instead of owning physical infrastructure. This eliminates the need for upfront hardware costs and supports scalability and cost efficiency, typically on a pay-as-you-use basis.

The **five fundamental characteristics** of cloud computing are:

- **On-Demand Self-Service:** Users can create and manage resources (like servers or storage) anytime without needing help from the cloud provider.
  - **Broad Network Access:** Cloud services are accessible from anywhere using standard devices like laptops, smartphones, and tablets.
  - **Resource Pooling:** The provider's computing resources are pooled to serve multiple customers (tenants) using a multi-tenant model, with different physical and virtual resources dynamically assigned and reassigned according to demand.
  - **Rapid Elasticity:** Resources can be quickly scaled up or down (provisioned and released) based on demand, often automatically, so users feel like they have unlimited capacity.
  - **Measured Service:** Cloud systems automatically control and optimize resource use by leveraging a metering capability. Users pay only for the resources they actually use.
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### 2. What are the different types of Clouds? Differentiate between them.

Cloud computing is categorized by its service model and its deployment model.

#### Service Models (Types of Service):

1. **Infrastructure as a Service (IaaS):** Provides access to fundamental IT infrastructure, such as virtual computers, storage, and networks. The user has control over operating systems and applications but doesn't manage the underlying physical infrastructure.
  - *Example: Amazon EC2*
2. **Platform as a Service (PaaS):** Provides a platform and environment for developers to build, test, and run applications. The provider manages the underlying servers and infrastructure, so developers can focus on writing code.

- *Example: AWS Elastic Beanstalk*

3. **Software as a Service (SaaS):** Provides fully developed software applications over the internet. The software is hosted and managed by the provider, and users access it via a web browser without needing to install or maintain anything.

- *Example: Google Docs*

#### **Differentiation between Service Models:**

| Aspect                  | IaaS                                  | PaaS                                       | SaaS                                  |
|-------------------------|---------------------------------------|--------------------------------------------|---------------------------------------|
| <b>What it Provides</b> | Virtualized hardware & infrastructure | Frameworks & platforms for development     | Fully developed software applications |
| <b>Management Focus</b> | Users manage VMs, OS, and storage     | Provider manages underlying infrastructure | Vendor manages everything             |
| <b>Customization</b>    | High (full infrastructure control)    | Moderate (limited to the platform)         | Minimal (limited to app settings)     |
| <b>Scalability</b>      | Scaled at the infrastructure level    | Scaled at the application level            | Scaled at the user/license level      |

### **3. Describe Cloud Architecture. What are the security issues in Cloud Computing?**

**Cloud Architecture** refers to the components and sub-components required for cloud computing. It typically consists of three main parts:

- **Front End:** The user interface and client devices used to access the cloud. This can be a "thin client" (like a web browser used in organizations) or a "fat/thick client" (like a home PC with advanced features for gaming).
- **Back-End Platforms:** The servers, storage systems, and other infrastructure that handle data processing, application management, and data storage.

- **Cloud-Based Delivery and Network:** The network that provides access to cloud services, using technologies like the Internet, Intranet, and Intercloud for connectivity and data transfer.

### Security and Privacy Issues in Cloud Computing:

1. **Data Confidentiality:** Ensuring that sensitive data is inaccessible to unauthorized users. A lack of trust in providers' ability to protect data is a major deterrent to cloud adoption.
2. **Data Loss:** The risk of data being lost or stolen, which is a major challenge for providers. Over 60% of users would avoid a vendor that has reported a data breach.
3. **Geographical Data Storage:** Data may be stored in a location outside the user's legal jurisdiction, raising concerns about legal accessibility and compliance with local laws.
4. **Multi-Tenancy Security:** Because resources are shared among multiple users, an attacker could exploit co-residence to attack other tenants.
5. **Transparency:** Providers must be open and clear about their security policies and practices. If policies are not easily understandable, transparency is poor.
6. **Hypervisor Issues:** The hypervisor manages multiple virtual machines (VMs) on one physical server. If an attacker compromises the hypervisor, they could potentially gain access to all the VMs on that server.
7. **Managerial Issues:** Non-technical problems like poor management, lack of control, or weak Service Level Agreements (SLAs) can create vulnerabilities, even if technical solutions are sound.

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### 4. Describe and Differentiate between Cloud Deployment Models.

A cloud deployment model defines where the cloud infrastructure is located, who owns and manages it, and who has access to it.

#### 1. Private Cloud:

- **Description:** Operated solely for a single organization. It can be managed internally or by a third party and can be located on-premises or off-premises. It offers the highest level of security and control.
- **Advantages:** Total control, elite security, and the ability to support legacy systems.
- **Disadvantages:** High cost and limited scalability due to dedicated hardware.

## 2. Public Cloud:

- **Description:** Infrastructure is owned, managed, and maintained by a third-party provider (e.g., AWS, Azure, GCP) and made available to the general public over the internet. Resources are shared among multiple tenants.
- **Advantages:** Minimal upfront investment, no setup or maintenance, and dynamic, near-limitless scalability.
- **Disadvantages:** Lower security due to shared resources and low customization.

## 3. Hybrid Cloud:

- **Description:** Combines a private cloud with one or more public clouds, allowing data and applications to be shared between them.
- **Advantages:** Ultimate flexibility (can keep sensitive data private and use public cloud for other tasks), cost-efficiency (using public cloud for traffic spikes), and targeted security.
- **Disadvantages:** Difficult to manage and complex, with potential for slow data transmission and latency.

### Differentiation between Deployment Models:

| Aspect                        | Public Cloud                          | Private Cloud                           | Hybrid Cloud                           |
|-------------------------------|---------------------------------------|-----------------------------------------|----------------------------------------|
| <b>Infrastructure</b>         | Shared among multiple organizations   | Dedicated to a single organization      | Customized mix of shared & dedicated   |
| <b>Cost Model</b>             | Pay-as-you-go; minimal upfront costs  | High initial investment & maintenance   | Variable; optimizes cost routing       |
| <b>Control &amp; Security</b> | Provider-managed; lower control       | Full internal control; highest security | Shared responsibility; complex routing |
| <b>Flexibility</b>            | High resource, low config flexibility | High config, restricted scaling         | Best of both worlds                    |

## Networking-1 Tentative Questions

### 1. What are the characteristics of LAN, MAN and WAN.

- **LAN (Local Area Network)**
  - **Scale:** Privately-owned network within a single building or campus, up to a few kilometers in size.
  - **Transmission:** High-speed data transfer, typically up to 10 Gbit/s.
  - **Management:** Easy to design and troubleshoot. All machines are often connected to a single cable.
  - **Topologies:** Uses Bus, Ring, Star, and Tree topologies.
  - **Advantages:** Fast data transfer and less expensive than other network types.
- **MAN (Metropolitan Area Network)**
  - **Scale:** Larger than a LAN but smaller than a WAN. It covers a city or a metropolitan area.
  - **Function:** Connects two or more LANs.
  - **Example:** A cable television network available in a city is a well-known example of a MAN.
- **WAN (Wide Area Network)**
  - **Scale:** Spans a large geographical area, often a country or a region. It links different metropolitan areas and national boundaries.
  - **Function:** Contains a collection of machines intended for running user (application) programs, called "hosts."
  - **Connectivity:** Communication is established using leased telephone lines, satellite links, and other long-haul channels.

### 2. What are the differences between LAN, MAN and WAN.

|                          |                                              |                                         |                                                     |
|--------------------------|----------------------------------------------|-----------------------------------------|-----------------------------------------------------|
| Feature                  | LAN (Local Area Network)                     | MAN (Metropolitan Area Network)         | WAN (Wide Area Network)                             |
| Geographical Area        | Small (e.g., a building, a campus)           | Medium (e.g., a city)                   | Large (e.g., a country, continent)                  |
| Speed                    | Very High (up to 10 Gbit/s)                  | High (slower than LAN, faster than WAN) | Lower than LAN and MAN                              |
| Ownership                | Typically private (owned by an organization) | Often shared or city-owned              | Typically private or leased from a telecom          |
| Cost                     | Least expensive to build and maintain        | More expensive than LAN                 | Most expensive due to long-distance links           |
| Design & Troubleshooting | Easy                                         | More complex than LAN                   | Most complex and difficult to troubleshoot          |
| Example                  | Network in a university building             | Cable TV network in a city              | The Internet, a network spanning multiple countries |

### 3. Describe the Network Edge and Transmission Media.

**Network Edge:** This refers to the computers and other devices that are at the edge (end) of a network. These are known as **hosts** or **end systems**. They are categorized into two types:

- **Clients:** Computer systems that request services or resources from a server. These are typically desktop PCs or workstations.
- **Servers:** Powerful computer systems that receive requests from clients, process them, and send a reply back.

The network edge includes two primary architectural models for connecting hosts:

- **Peer-to-Peer Network:** A group of computers connected together with no central server. Each computer acts as both a client and a server, sharing resources directly with each other.
- **Client/Server Network:** A system where clients connect to a central, powerful server to access resources. The server provides services like running applications, managing files, or handling print requests. All files and resources are centralized on the server.

**Transmission Media:** This is the physical or non-physical pathway that carries information (data) from a source to a destination. It is classified into two main categories:

- **Guided Transmission Media (Physical/Bounded):** Provides a physical channel for data to travel along a solid medium.
    1. **Twisted Pair:** Two insulated copper wires twisted together. The twisting helps cancel out electromagnetic interference. Used in telephone networks and Ethernet cables.
    2. **Coaxial Cable:** Has better shielding than twisted pair, allowing for longer distances and higher speeds. Consists of a copper core, insulating material, a braided outer conductor, and a plastic sheath. Used for cable TV and internet.
    3. **Fiber Optics:** Uses glass or plastic fibers to transmit data as pulses of light. It offers massive bandwidth, very high speeds, and immunity to electromagnetic interference. It is the backbone of modern high-speed networks.
  - **Unguided Transmission Media (Wireless/Boundless):** Transports electromagnetic waves through free space without a physical conductor.
    1. **Radio Transmission:** Easy to generate, travels long distances, and penetrates buildings. It is omnidirectional, so a clear line-of-sight is not required.
    2. **Microwave Transmission:** Travels in a straight line and requires a clear line-of-sight. It uses parabolic antennas to focus the signal and is used for point-to-point communication like satellite links.
    3. **Infrared:** Used for short-range communication (e.g., remote controls). It is directional, cheap, and secure, but cannot penetrate solid objects.
    4. **Lightwave Transmission:** Uses laser light to transmit data through the air. It offers high bandwidth but is susceptible to atmospheric interference like rain and fog.
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## Networking-2 Tentative Questions

### 1. How data flows in OSI Model (Mainly read the examples)

When data is transferred from one device to another, it travels through the 7 layers of the OSI model. The process follows these steps:

#### 1. Sender Side (Data Travels Down):

- **Application Layer:** The user's application (e.g., email client) creates the data.
- **Presentation Layer:** The data is formatted and encrypted for transmission.
- **Session Layer:** A connection is established and managed between the sender and receiver.
- **Transport Layer:** The data is broken down into smaller segments, and sequence numbers and error-checking info are added for reliable delivery.
- **Network Layer:** The segments are packaged into packets with logical addressing (IP addresses) to find the best route.
- **Data Link Layer:** The packets are framed (encapsulated into frames), and hardware (MAC) addresses are added for the next device on the local network.
- **Physical Layer:** The frames are converted into bits (0s and 1s) and transmitted as electrical, optical, or radio signals over the physical medium.

2. **Receiver Side (Data Travels Up):** The process is reversed in the reverse order. Each layer on the receiving end strips off the header added by its corresponding layer on the sending end. The Physical Layer receives the signals and converts them back into bits, and the Data Link Layer checks for errors. Finally, the data is reassembled and presented to the receiving application.

#### Example: Person A sends an e-mail to Person B

- **Application:** Person A writes the email in their email application (e.g., Gmail).
- **Presentation:** The email data is formatted and encrypted for secure transmission.
- **Session:** A connection is established between Person A's computer and the mail server.
- **Transport:** The email is broken into smaller segments, and sequence numbers are added to ensure it arrives in the correct order.
- **Network:** Each segment is placed into a packet with an IP address to route it across the internet to the correct destination.

- **Data Link:** The packets are placed into frames, and MAC addresses are added so the data can be delivered to the next device on the local network (like a router).
- **Physical:** The frames are converted into electrical/optical signals and sent over the network.

The process reverses at Person B's end, with the layers stripping off the added information until the email is decrypted and displayed in Person B's inbox.

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## 2. What are the differences between TCP/IP and OSI Model. Why TCP/IP model is better than OSI model.

### Differences between the OSI and TCP/IP Models:

| Basis of Comparison                      | OSI Model                                                                              | TCP/IP Model                                                                              |
|------------------------------------------|----------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------|
| <b>Number of Layers</b>                  | 7 Layers (Physical, Data Link, Network, Transport, Session, Presentation, Application) | 4 Layers (Network Access, Internet, Transport, Application). Sometimes shown as 5 layers. |
| <b>Nature</b>                            | Conceptual / Theoretical reference model                                               | Protocol-driven; based on real, working protocols                                         |
| <b>Development</b>                       | Developed first, then protocols were fitted to it                                      | Protocols (TCP, IP) came first; the model was built to describe them                      |
| <b>Session &amp; Presentation Layers</b> | Separate layers                                                                        | Not present — their functions are merged into the Application layer                       |
| <b>Layer Boundaries</b>                  | Strict; each layer has a clearly defined function                                      | Loosely defined; less rigid separation                                                    |

|                       |                                                |                                                               |
|-----------------------|------------------------------------------------|---------------------------------------------------------------|
| Basis of Comparison   | OSI Model                                      | TCP/IP Model                                                  |
| Usage Today           | Used mainly for teaching and design reference  | Actually used to run the Internet and real-world networks     |
| Reliability Guarantee | The model does not guarantee reliable delivery | Guarantees reliable delivery at the Transport layer (via TCP) |

### Why TCP/IP is Considered Better/More Popular than OSI:

1. **Simpler Structure:** TCP/IP has only 4 layers compared to OSI's 7, making it easier to implement, understand, and troubleshoot in real systems.
2. **Protocol-Driven Design:** TCP/IP was built from working protocols (TCP, IP) that were already in use. OSI was primarily a theoretical framework designed first.
3. **Flexibility and Robustness:** TCP/IP adapts well to different hardware and network environments and includes built-in error handling, routing, and congestion control.
4. **Open Standard:** TCP/IP is open, free to use, and not controlled by any single organization. This "openness" helped it gain universal acceptance and adoption.
5. **Actual Use vs. Conceptual Model:** OSI is a great model for learning the principles of networking, but TCP/IP is the practical model that actually runs the Internet and is used in all modern networking.